Reconciling Top8000 Ramachandran data with the Richardson Laboratory `rama8000` reference distributions

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15 October 2026

Background

The MolProbity Ramachandran criteria [1–3] were updated from the Top8000, a set of about 8000 high-resolution, quality-filtered protein chains, using six residue categories (General, Gly, Ile/Val, pre-Pro, trans-Pro, cis-Pro) [4–6]. The chains were selected in 2011 [7, 8]; later reference sets were used for rotamers [9, 10] and for the Top2018 best-parts dataset [11, 12]. Cis peptides are discussed in [13]; other non-parametric estimates of Ramachandran distributions are described in [14, 15].

This note reconciles Top8000 Ramachandran data, as we derive them from the laboratory's 2011 chain files, with the Richardson Laboratory's `rama8000` distributions, at the level of category counts and contour polylines. Our published Top8000 dataset is [16].

In brief

A.1 Data. `top8000_chains_70.tgz`; `Top8000_Rama_2013.zip`; `rama8000` NDFT tables; contour Makefile.

A.2 Programs. Silk; kin2Dcont; Dangle and Ramalyze (2011 versions) as cross-checks of our Python parser.

B. Implementation. Revised Ramachandran category precedence, so that pre-Pro Ile/Val go into pre-Pro, not Ile/Val; the original Top8000 model files used for atom records; a revised rule set that filters on deposited heavy-atom model values only.

1 Required materials

These are the original Richardson Laboratory files and programs the comparison needed.

Data files

1. `top8000_chains_70.tgz` [17]: 7,957 Top8000 chain files from 2011, with Reduce 3.15 hydrogens [18, 19]. Source: the laboratory website, recovered via the Internet Archive. Used as the input for our category counts; the hydrogens are ignored.
2. `Top8000_Rama_2013.zip` [20]: the laboratory's filtered residue lists (one per category) plus `get_srcdata.py`, `filters_mysql.txt`, `symmetrize.awk`, the Silk Makefile and Top8000 Origins [7]. Source: laboratory archive. Used as the laboratory counts we compared against, and as the source of the filtering rules (the SQL in `get_srcdata.py`).

3. `rama8000` NDFT tables (`*.ndft`): `rlabduke/javadev`, commit `24e0666` [21]. Used as the laboratory grids in the contour comparison, and to confirm that the laboratory's filtered residue lists reproduce them exactly.
4. Contour Makefile: `rlabduke/reference_data`, commit `68f581a` [22]. Used for the kin2Dcont options and contour levels.

Programs

5. Silk (two-pass kernel density estimation): `rlabduke/javadev`, commit `24e0666`, rebuilt [21].
6. kin2Dcont (contour polylines): `rlabduke/kincontours`, commit `3da22b4`, rebuilt [23].
7. Dangle 1.07 and Ramalyze, mid-2011 versions: `rlabduke/javadev`, commit `eb1c7f48` (2011-06-16), rebuilt [21]. Cross-checks only; no reported number depends on them.

Not needed. Reduce and MolProbity. Residue filtering uses deposited heavy-atom model values only; Reduce hydrogens and flips matter for rotamers, not for Ramachandran filtering.

Not available. The laboratory's 2011–2013 MySQL database (not preserved); `cont.php`, the contour post-processing script (unpublished).

Our own code. A Python parser and filters for ϕ/ψ, ω and the per-residue values; ContourPy [24] for the marching-squares check.

2 Comparison

2.1 Setup

Chain data	(a) The laboratory's 2011 Top8000 chain files (`top8000_chains_70.tgz`; 7,957 single chains). (b) Our current RCSB downloads of the same 7,957 chains, as used for our published dataset [16]. Hydrogens are ignored in both.
Program	Our own Python parser (ϕ, ψ, ω and per-residue filters). The 2011 Java Dangle and Ramalyze were run as cross-checks; their ϕ/ψ agree with ours.
Laboratory reference	The laboratory's filtered residue lists (`get_srcdata.py` output). Silk run on these lists reproduces all six `rama8000` NDFT tables exactly.

2.2 Chain data: the laboratory's 2011 files and our downloads

Both start from the chain list `Top8000-best_hom70_pdb_chain_list.csv` [22] (7,957 chains from 7,737 entries); our copy is byte-identical to the laboratory's. The same list underlies the published `rama8000` contours: the laboratory's filtered residue lists draw on 7,810 of these chains, 898 of which are not in `Top8000-SFbest_hom70`.

All 7,957 chains (7,737 entries) and their chain IDs agree with the Richardson laboratory's, and we could download every entry, including three now in the obsolete archive (3m8t, 3n22, 3r4e). For the mainchain atoms N, CA, C and O, the laboratory's files contain 7,550,002 records and ours 7,550,019. Matching records by chain, residue number, insertion code, atom name and altloc, 7,542,952 are identical in residue name, coordinates, occupancy and B. Another 6,150 are matched but differ in at least one field: residue name 3,264, B 5,605, occupancy 396, coordinates 4,812, every

coordinate change exceeding 1 Å. A record can differ in more than one field. A further 900 records appear only in the laboratory's files and 917 only in ours, from renumbered, added or removed residues. All 7,967 non-identical records lie in 64 of the 7,737 entries; 8 re-refined or re-sequenced entries (3me5, 3hkx, 3kda, 3oon, 2pvq, 3h87, 2oyp, 1gug) account for 7,339 of them.

Filtering rules. All tests use deposited heavy-atom model values only.

1. Peptide-linked on both sides (C–N < 2.0 Å).
2. Own mainchain (N, CA, C, O): $1 < B < 30$ (the SQL's `1 < max_mc_b < 30`); for MSE, $B < 30$ over all heavy atoms.
3. Every heavy atom of the residue at occupancy 1.
4. Non-Pro residues: no alternate location on the preceding residue's C, so that ϕ is single-valued.
5. ω defined; trans $|\omega| \geq 150°$, except cis-Pro $|\omega| < 30°$.
6. Standard 20 residues + MSE.
7. General, Ile/Val, pre-Pro only: next residue is standard 20 or MSE.

Rules 3 and 4 are our heavy-atom reading of the SQL's `min_res_q = 1`; how that column was computed is not preserved.

2.3 Category counts

	Richardson	Ours, 2011 files		Ours, current files	
Category	laboratory	count	Δ	count	Δ
General	1,061,639	1,061,572	−67	1,061,551	−88
Gly	116,589	116,579	−10	116,573	−16
Ile/Val	195,083	195,080	−3	195,083	0
pre-Pro	61,568	61,574	+6	61,573	+5
trans-Pro	66,364	66,364	0	66,362	−2
cis-Pro	3,654	3,654	0	3,654	0
Total	1,504,897	1,504,823	−74	1,504,796	−101

On the 2011 files, 90 residues are only in ours and 164 only in the laboratory's lists. On the current files, 266 are only in ours, 367 only in the laboratory's lists, and 307 have the same residue number but a different category.

2.4 Explaining the differences on current files

Each of the 940 residues that differ on the current files is assigned one cause. If the same rules on the 2011 files also disagree with the laboratory, it is a rule difference. Otherwise the 2011 file reproduces the laboratory's choice and the cause is the change between the two file versions.

Cause	Residues	Entries
File differences (entries revised since 2011)	**686**	
Residue name changed (re-sequenced)	453	3me5, 3hkx, 3h87, 2oyp, +4
Residue number absent in one file	160	3oon, 3hkx, 2oyp, 3h87, 3me5
Mainchain *B* changed	50	2pvq (34), 3hkx, 3me5, +2
Occupancy, altloc, ω, category, next residue	23	1–5 residues each
Rule differences (also on the 2011 files)	**254**	157 entries, at most 14 each

The 254 rule differences are not explained by any heavy-atom rule we have. They include 74 non-Pro residues that the laboratory kept although the preceding C has an alternate location (rule 4), out of 10,395 such residues that otherwise pass:

Preceding C	Laboratory kept	Laboratory dropped
Single altloc record	37	56
Alternates at the same position	2	43
Alternates at different positions	35	10,222

A rule we invented and do not adopt. Replacing rules 3–4 with "every atom in the 2011 file at occupancy 1", which includes the hydrogens added by Reduce 3.15, improves the agreement on the 2011 files to +5 (95 only in ours, 90 only in the laboratory's lists); on the current files, with Reduce 4.16 hydrogens, it gives −954. This rule has no support in the recollections of the original researchers or in the original filtering artifacts we have, and it goes against the principle of filtering only on experimentally determined heavy-atom positions. We report it only as a comparison.

2.5 Contour polylines

Contours were drawn with the laboratory's own program, kin2Dcont [23], using the options in the laboratory's Makefile [22], at the Ramalyze levels [21]: inner 98% for all classes; outer 99.95% General, 99.80% cis-Pro, 99.90% others.

Laboratory NDFT grid vs our Silk grid, both contoured with kin2Dcont: every contour has the same number of polylines. The largest deviation is 0.73°; mean deviations are 0.00–0.07°.

For comparison, the 1° sampling offset between the laboratory's contour grid and the NDFT grid moves lines by up to 0.88°, and changing the contouring method moves them by up to 0.42°.

(i) Density estimation: Silk two-pass adaptive kernel density estimate

	Original (Richardson laboratory)	Ours
Program	Silk (`rlabduke/javadev`)	Silk, javadev `24e0666`, rebuilt (OpenJDK 21)
Input	laboratory's filtered residue lists	our ϕ/ψ under the filtering rules
Gly	mirrored: (ϕ, ψ) and $(-\phi, -\psi)$	same, mirrored explicitly
Kernel, 1st pass	cosine, half-width 5° (`-cosine=5`)	same
2nd pass	adaptive, `-twopass=6.5` (General), 8 (all others)	same
Grid	2° (`-gridsize=2`), periodic (`-wrap`)	same
Output	`-fraction` (fraction of data at lower density)	same
Grid placement	NDFT tables: bins centred on odd degrees (`-bounds=-180,180,-180,180`); contour kinemages: on even degrees (`-bounds=-181,179,-181,179`)	both; contours compared on the odd-degree grid

Check: Silk with the original options on the laboratory's filtered residue lists reproduces all six published `rama8000` NDFT tables bin for bin.

(ii) Contouring

Original. kin2Dcont (`rlabduke/kincontours`, J. M. Word [23]), options

```
-samp -last -nosmooth -group -wrap -180 180 -180 180 -g2
```

Method: Zyda contour tree [25], not marching squares.

- Each grid value sits on a lattice node; missing samples count as 0.
- The wrap edge is duplicated for periodicity.
- Each 2° cell is split into four triangles around a centre point (the mean of its four corners).
- Contour crossings are linearly interpolated along triangle edges.

Levels (fraction): inner 98% = 0.02 for all classes; outer 99.95% = 0.0005 General, 99.90% = 0.001 others, 99.80% = 0.002 cis-Pro. These are the outlier cutoffs in the laboratory's Ramalyze (`Ramachandran.java`; cis-Pro: `CISPRO_ALLOWED`) [21]; all are in the Makefile's level list. The laboratory's output then went through an unpublished script (`cont.php`).

Ours.

- kin2Dcont, `rlabduke/kincontours 3da22b4`, rebuilt with the original options. On the odd-degree grid: `-wrap -179 181 -179 181`.
- Marching squares as a method check: ContourPy [24], applied to the same grid padded periodically by 4 cells.

(iii) Results

Laboratory NDFT grid vs our Silk grid, both contoured with kin2Dcont: every contour has the same number of polylines. The largest deviation is 0.73° (Ile/Val outer); mean deviations are 0.00–0.07°.

For comparison:

- the 1° offset between the even- and odd-degree grids moves lines by up to 0.88° (cis-Pro outer, the sparsest class);
- kin2Dcont vs marching squares on the same grid differ by up to 0.42°.

Note: the contour comparison used our current RCSB files under the earlier rule set, not the 2011 files and final rules above. So the contour lines are largely insensitive to modest variation in the residue data set (about 0.1–0.2%), to the 1° grid offset, and to the contouring method (kin2Dcont vs marching squares). The density estimate itself was not varied: both sides used the same Silk program and parameters.

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